

Word2Vec Numerical Notes

Mastering Word Embeddings: CBOW & Skip-gram Architecture

Fundamental Concepts

Cosine Similarity: $\cos(\theta) = (A \cdot B) / (||A|| \cdot ||B||)$

Softmax Probability: $P(w_o | w_I) = \exp(v'_{w_o} v_{w_I}) / \sum \exp(v'_w v_{w_I})$

Sigmoid Function: $\sigma(z) = 1 / (1 + \exp(-z))$

Section 1: Basic Level

Question 1: Cosine Similarity Calculation

Given vector $\mathbf{v}_1 = [0.4, 0.9]$ (King) and $\mathbf{v}_2 = [0.3, 0.8]$ (Queen). Calculate the cosine similarity.

Step-by-Step Solution:

- Dot Product: $(0.4 * 0.3) + (0.9 * 0.8) = 0.12 + 0.72 = 0.84$
- Magnitude v_1 : $\sqrt{(0.4^2 + 0.9^2)} = \sqrt{(0.97)} \approx 0.9849$
- Magnitude v_2 : $\sqrt{(0.3^2 + 0.8^2)} = \sqrt{(0.73)} \approx 0.8544$
- Cosine Similarity: $0.84 / (0.9849 * 0.8544) \approx 0.998$

Answer: Similarity ≈ 0.998

Question 2: Skip-gram Training Pairs

Sentence: "The cat sat on the mat". Window size $k = 2$. Target word: "sat". List the training pairs.

Step-by-Step Solution:

1. Identify Context (2 left, 2 right): "The", "cat", "on", "the"
2. Map Target to each Context word.

Answer: (sat, The), (sat, cat), (sat, on), (sat, the)

Section 2: Advanced Level

Question 3: Softmax Probability

Center vector $\mathbf{v}_{\text{apple}} = [1, 0]$. Vocabulary output vectors: apple $[0.5, 0.5]$, orange $[1, 1]$, juice $[0, 1]$. Calculate $P(\text{orange} \mid \text{apple})$.

Step-by-Step Solution:

1. Dot Products: $\text{Score}(\text{apple})=0.5$, $\text{Score}(\text{orange})=1.0$, $\text{Score}(\text{juice})=0$
2. Exponents: $e^{0.5} \approx 1.649$, $e^{1.0} \approx 2.718$, $e^0 = 1.0$
3. Sum of Exponents: $1.649 + 2.718 + 1.0 = 5.367$
4. Probability: $2.718 / 5.367 \approx 0.506$

Answer: $P(\text{orange} \mid \text{apple}) \approx 0.51$

Question 4: Negative Sampling Scores

Calculate sigmoid scores for a positive pair (dot product 2.0) and a negative pair (dot product -3.0).

Step-by-Step Solution:

1. Positive Sigmoid: $1 / (1 + e^{-2.0}) \approx 0.88$

2. Negative Sigmoid: $1 / (1 + e^{3.0}) \approx 0.047$

Answer: Positive ≈ 0.88 , Negative ≈ 0.05